CDEQ V2.0: The Unifying Master Equation for Constructive Lyapunov Stability

构造性李雅普诺夫稳定·大一统主方程

以大一统公式，闭合李雅普诺夫稳定百年构造性缺口

- CDEQ V2.0 唯一官方主方程

- 全体系收敛于一行，不可删减、不可替换、不可变形

- 理论根基：PCTT-CW 五元顺时针循环拓扑

- 工程实现：FPGA 可综合硬核 Verilog

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Abstract

Since 1892, Lyapunov stability has formed the theoretical foundation of system stability, but a constructive, deterministic, finite-time, hardware-synthesizable solution remained missing. This paper presents CDEQ V2.0, a unified framework expressed in one unique, canonical master equation. The equation integrates cyclic dynamical foundations (PCTT-CW), invariant structural constraints, and global finite-time convergence verification. A fixed Verilog implementation is provided as the canonical hardware realization. This work closes the 134-year implementation gap in Lyapunov stability.

Keywords: Lyapunov stability; constructive control; discrete dynamical systems; finite-time convergence; canonical equation

1. Definition

A discrete-time system is constructively stable if:

1. States are uniformly bounded.
2. Update rule is fixed and deterministic.
3. Convergence is achieved in provable finite time.
4. The system is directly synthesizable in hardware.

2. Foundational Basis: PCTT-CW

The dynamical core of CDEQ is the PCTT-CW pentacyclic clockwise topology:
- Five-node fixed cyclic structure.
- Adjacent nodes reinforce; interval nodes restrain.
- Golden-ratio coupling $\varphi=0.618$, $\bar{\varphi}=0.382$.
- Deterministic, bounded, provably stable.

3. Canonical Master Equation (Unique Fixed Form)

The entire CDEQ framework is defined by one single, immutable formula:
$\Psi' = \mathrm{clip}[\varphi\Psi_{-1} - \bar{\varphi}\Psi_{-2} + \omega\xi] + \nabla C \rightarrow \Phi<\Gamma \ \& \ |E_{11}-5|<\varepsilon \ \& \ |\Sigma-15|<\varepsilon \ \forall 10$
This is the only official, canonical form of CDEQ V2.0.
No modification, substitution, or simplification is permitted.

4. Framework and Contributions

The master equation unifies:

1. PCTT-CW cyclic dynamics
2. Invariant center $E_{11}=5$ and sum $\Sigma=15$ constraints
3. Global finite-time convergence ($\forall 10$ steps)

Key contributions:

- Unique closed-form expression
- Fixed structure, no tuning parameters
- Finite-time convergence guarantee
- Directly synthesizable on FPGA/ASIC

5. Canonical Hardware Implementation

plaintext
```
module pctt_cw();
real P[0:4],A[0:4];
real S=0.618,C=0.382;
integer PREV[0:4]='{4,0,1,2,3};
integer PREV2[0:4]='{3,4,0,1,2};

initial begin
P[0]=0.5;P[1]=0.5;P[2]=0.5;P[3]=0.5;P[4]=0.5;
```

```
end

task update;
integer i;
for(i=0;i<5;i=i+1) begin
A[i]=S*P[PREV[i]]-C*P[PREV2[i]];
P[i]=P[i]+A[i];
P[i]=P[i]>1.0?1.0:P[i]<0.0?0.0:P[i];
end
endtask

endmodule
```

6. Conclusion

CDEQ V2.0 provides a unified, constructive, hardware-ready solution to the Lyapunov stability implementation gap. Built on the PCTT-CW foundation, the unique canonical master equation establishes a complete, minimal, and formally verifiable framework for global stability and finite-time convergence.

References

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